Implementation Plan: live within 90 days

PA Copilot · Bellcourt Health Administrators · Ratnesh Singhaniya · 5 October 2026

Two dates drive the plan. (1) Riverbend requires **60 days' written notice** before any automated or AI-assisted tool enters the PA workflow (Addendum §5), so the notice goes out on **day 1** and Riverbend go-live is **day 61 or later**. (2) Riverbend can put the contract out to bid if Q3 doesn't improve (CEO, LP p.23), and it can terminate for cause if the CAP isn't met within 120 days (Addendum §7). Self-funded employers are not covered by the Riverbend notice, so they go live first and generate real-world evidence.

Phase	Work	Owner(s)	Exit gate
0 · Days 1–10 Set up	Send the 60-day notice to Riverbend, with this design and the test results. Security review: PHI minimization, role-based access, logging (Paul's requirements). Stand up in Bellcourt's Azure tenant (HIPAA BAA): container app, Azure OpenAI, Key Vault, Log Analytics. Read-only access to the PACE replica, fax share, portal webhook and eligibility files. Name a Library owner.	FDE, IT (Paul), Security, CMO's office	Notice sent; access granted; security sign-off
1 · Days 11–30 Build + shadow	Index the full UM Library (~1,100 files). Clinical Policy clears the needs-review list (duplicates, undated files, conflicts). Extend the service keyword catalogue to every service and all 38 employer plans. Point the LLM client at Azure OpenAI (one setting). Run in **shadow mode** on the live queue: the tool runs, staff don't see it, and its output is compared with nurse decisions and QA.	FDE, Clinical Policy, IT	Index ≥ 99% of active documents classified; shadow run on ≥ 500 live cases
2 · Days 31–60 Validate + self-funded go-live	Validate on a fresh QA sample of ≥ 100 cases: correct source shown ≥ 95%, every quote verified or flagged, 0 automated adverse actions. Train intake (30 min) and nurses (45 min) with 2 super-users. **Go live for self-funded employers.** Share shadow results and sample audit records with Riverbend (Linda Cho). Add the Texas AI-disclosure line to Riverbend notice templates.	FDE, UM Ops (Diane Mercer), QA lead, super-users	Thresholds met; Riverbend reviewed the evidence
3 · Days 61–90 Riverbend go-live	**Go live for Riverbend MA and Marketplace** once the notice period ends. Route AZ MA cases needing a denial to AZ-licensed MDs early. Weekly scorecard: on-time from receipt, hours pending for info, accuracy (QA), overturn rate, guardrail interventions. Weekly fix loop. Retire UM-MEMO-2025-19 through the Clinical Policy Committee.	UM Ops, FDE, QA	Q4 RB-MA on-time trending to ≥ 97%; scorecard sent to Riverbend

Go / no-go rules

Gate	Go if	Otherwise
Shadow → self-funded live	Correct source ≥ 95% on the fresh QA sample; quote verification 100% (verified or flagged); 0 decision language reaching staff	Fix and re-run shadow for 2 weeks
Self-funded → Riverbend live	Notice period ended; Riverbend has reviewed results; no open security findings	Stay self-funded only; keep sharing evidence
Any time	—	Kill switch: the tool is read-only, so turning it off returns staff to today's process with no data loss

Team and effort

FDE (full time) · 1 IT engineer from Paul's team (~0.3 FTE, access and hosting) · Clinical Policy / Library owner (~0.2 FTE) · UM Ops champion (Diane Mercer) · 2 super-user nurses · QA lead. No new nurse headcount is assumed. The metric to watch is the share of 14.2 minutes per case spent locating and reading rules (re-measured in a repeat time-and-motion study at day 75).

Design Note: technology choices

One principle: **use code wherever the answer is a rule; use AI only where messy text must be read; verify the AI with code.**

Need	Choice	Why (and what I rejected)
Pick the governing documents	Plain Python rules over a metadata index	GOV-01 is a fixed ranking, and the right policy version is a date comparison. Rules give the same answer every time and explain every rejection ("v1 not used: effective to 2025-12-31"). *Rejected: vector search.* MP-101 v1 and v2 read almost alike, so "most similar" is not "in force on the date of service". Also rejected: a vector database, which is unnecessary when the rules narrow each case to 2–4 short documents.
Metadata index	Parser reading each document's own header; JSON file (demo) → Azure SQL / Blob (production)	The documents already state ID, version, status and effective dates. Unreadable, duplicate or conflicting files go to a needs-review list, never into use.
PDF text	`pdfplumber`, table-aware	Plan exclusions, the Addendum §2.1 Medicare table and the PACE screen schedule are tables; plain text extraction mixed their columns.
Read faxes	Tesseract OCR, local + label parser	Fax images are full of PHI and stay on Bellcourt machines. All 13 test faxes read correctly. *Rejected: sending images to a vision model* (PHI exposure, less predictable, could follow text inside the image).
Read clinical notes against criteria	LLM via an OpenAI-compatible API: Claude Sonnet 4.5 on OpenRouter (demo) → Azure OpenAI in Bellcourt's tenant (production)	The one step that needs language understanding. Swapping provider is one setting (base URL + key). Production stays inside the BAA-covered tenant approved for PHI (LP p.21). The demo used synthetic data only. Temperature 0; answers saved by fingerprint, so the same input gives the same answer at no extra cost (the whole evaluation, ~59 reviews, ran within a $5 budget).
Approach	**RAG with rule-based retrieval** in a fixed pipeline	*Rejected: agentic RAG and autonomous agents.* There is nothing to plan: steps are identical per case. An agent that can act could pend, delay or deny, which Addendum §5 forbids. Fixed steps are auditable.
Trust the AI output	Guardrails in code	Output has no decision field. Every quote is matched word-for-word against the cited document and every piece of evidence against the notes. Verdict sentences are removed (not words: "recommended by a spine surgeon" is policy text). OR-alternatives are read as one group. Identity and planted instructions are removed before the call.
User interface	Streamlit (demo) → internal web app or PACE side panel	Python end to end, fast to build, enough for intake and nurse views. A static export (`demo/index.html`) lets anyone view results without setup.
Audit and testing	JSON audit record per run; `pytest` (34 tests); evaluation scripts over the QA audit	Addendum §5 requires inputs and outputs on audit. Every trap found in the data is a permanent test.

What I'd change at scale: move the index and audit log to Azure SQL; get the keyword catalogue from PACE service-to-policy mappings instead of maintaining it by hand; add a reviewer-feedback button feeding QA; and serve the same rules engine through the January 2027 electronic PA API and a provider status page.